Yugabharathi Jaisankar

R&D / Software Engineer — Game Systems, Automation & Tooling

Erode, Tamil Nadu, India | +91 8838144028 | yuga.bharathijai2106@gmail.com

github.com/spiicez21 | github.com/SPiceZ21 (SPZ-Core org) | linkedin.com/in/yugabharathi21 | yugabharathi21.netlify.app

SUMMARY

Computer Science engineering student (B.E., graduating 2027) who architects and maintains SPZ-Core, an open-source racing framework for the FiveM game platform spanning 35+ repositories: server-authoritative race engine, powertrain simulation, versioned database migrations and one-command deployment. Debugs runtime faults from player-facing symptoms back to root cause and builds tooling that removes manual work. Seeking an R&D role in automation frameworks and game test tooling.

TECHNICAL SKILLS

Languages: C++, C, Lua 5.4, JavaScript, TypeScript, Python, SQL, C#;
Core CS: Object-oriented and modular design, data structures and algorithms, pointers and memory management, event-driven architecture, software development lifecycle (SDLC)
Debugging & Troubleshooting: Root-cause analysis, step-through debugging (Visual Studio / VS Code / GDB), error relay and log analysis, runtime diagnostics, performance profiling
Testing, Automation & Deployment: Versioned migrations with boot-time gating, one-command server provisioning (txAdmin), scripted validation utilities, regression checking, automated build checks (GitHub Actions), CI/CD fundamentals
Version Control: Git, GitHub, multi-repository organisation management, branching, pull requests, maintainer code review, issue triage, semantic versioning
Game Systems: FiveM / CitizenFX, server-authoritative state, routing buckets and session lifecycle, sector timing and telemetry, ghost/replay data, powertrain simulation, HUD/UI
Application & Data: React.js, Next.js, Node.js, Express.js, MySQL, MongoDB, REST APIs | **Tools:** Visual Studio, VS Code, Docker, Postman, Figma, Blender
Currently strengthening: Jenkins and TeamCity pipelines, C# tooling, automated game-test framework design
Behavioural: Problem solving | Teamwork | Adaptability | Communication | Initiative taking | Quick learner, open to feedback in fast-paced iterative environments

PROJECTS & OPEN-SOURCE EXPERIENCE

SPZ-Core — Modular Racing Framework for FiveM | Architect & Maintainer (GPL-3.0) 2025 – present

- Designed and maintain a racing-only game framework across 35+ interoperating repositories (Lua, JavaScript, TypeScript), with a bootstrap module that solely owns sessions, routing buckets, permissions and config — every module goes through its exports, never shared state.
- Built a server-authoritative race engine (queue, grid, sector timing, live splits) so the server, not the client, decides every outcome that affects a race result.
- Implemented a vehicle powertrain simulation — torque curves, gearing, clutch, turbo and LSD behaviour — driving performance-point class bands used for race balancing.
- Made schema evolution safe: numbered SQL migrations apply once at boot, record themselves in a migrations table, and gate the framework-ready event so no module can query a half-built database.
- Added live-environment diagnostics — client error relay into the server log, runtime status and debug commands — plus a one-command txAdmin recipe that provisions a full server reproducibly.

SPZ-HUD-FiveM — Game HUD / UI Module (8 GitHub stars) 2024 – 2025

- Rebuilt the Qbox HUD with a responsive, higher-readability interface and reduced per-frame update cost so it ran cleanly alongside other resources in live sessions.
- Debugged client–server event flow, reproducing player-reported faults and tracing them to the responsible resource; documented setup for other operators.

SimpMusic — Desktop Mini Player Feature (Open-Source Contribution, PR #1614) 2025

- Navigated an unfamiliar production codebase to locate the playback and window management layers, implemented a floating mini player with synchronised UI state and low runtime overhead, and iterated through maintainer review; shipped in v1.0.2.

GoFr — Summer of Code 2025 Contributor Jun – Aug 2025

- Contributed to an open-source Go project over three months, working against the project's automated CI checks, linting and review standards.

EDUCATION

B.E. Computer Science and Engineering — Kongu Engineering College, Perundurai, Erode 2023 – 2027 (expected)
Coursework: Data Structures and Algorithms (C/C++), Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Software Engineering

ACHIEVEMENTS & CERTIFICATIONS

2nd Prize — Hackbuzz 2k25 Inter-Department Hackathon | **Participant** — Adobe India Hackathon, Round 1 | 60+ public repositories maintained | **Certification** — Oracle APEX Cloud | Also built: real-time typing metrics engine and ThinkForge AI study assistant (48-hour hackathon).